

26高数强化 (18)

18	二重积分举例（二重积分计算、累次积分、综合题及不等式）	P191-P203
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下 p204~216

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还不关注，
你就慢了



@武忠祥老师 @考研可爱因子

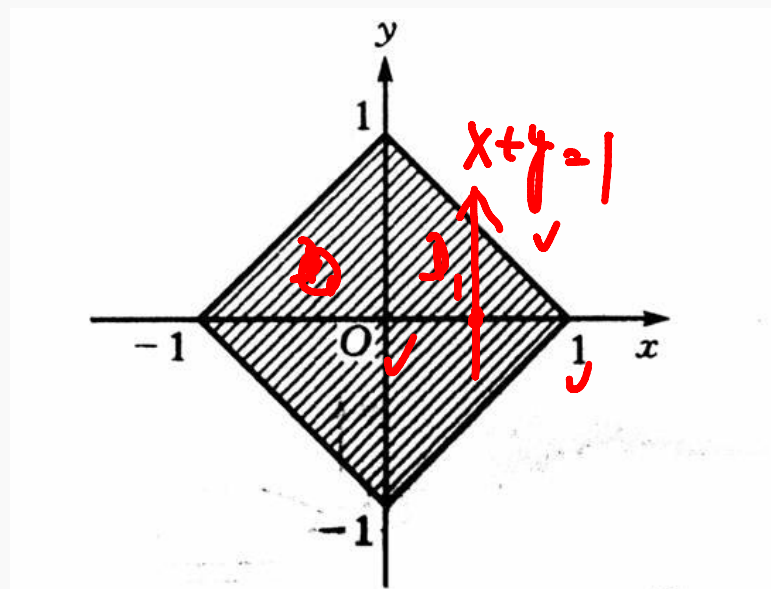


题型一 计算二重积分

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【例1】计算 $\iint_D [|xy| + \sin(xy^2)] d\sigma$, 其中 D 由曲线 $|x| + |y| = 1$ 所围成.

【解】 原式 $= \iint_D |xy| d\sigma = 4 \iint_{D_1} xy d\sigma$
 $= 4 \int_0^1 dx \int_0^{1-x} xy dy = \frac{1}{6}.$



还不关注，
你就慢了



【例2】设区域 D 为 $x^2 + y^2 \leq R^2$, 则 $\iint_D (\frac{x^2}{a^2} + \frac{y^2}{b^2}) d\sigma = \underline{\hspace{2cm}}$.

【解1】 $\iint_D (\frac{x^2}{a^2} + \frac{y^2}{b^2}) d\sigma = \int_0^{2\pi} d\theta \int_0^R (\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}) \underline{r^3} dr = \frac{\pi R^4}{4} (\frac{1}{a^2} + \frac{1}{b^2})$

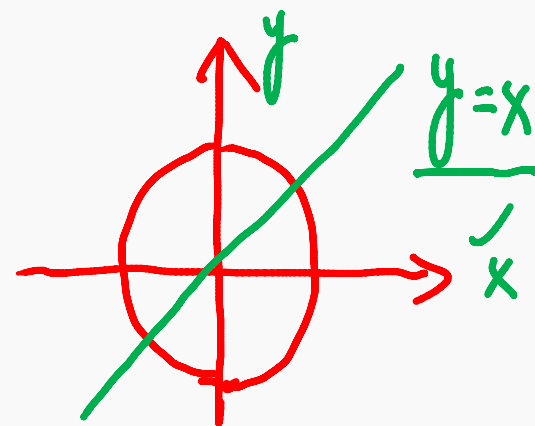
Handwritten note: $\int_0^{2\pi} \cos^2 \theta d\theta = \int_0^{2\pi} \sin^2 \theta d\theta = \pi$

【解2】 $\iint_D (\frac{x^2}{a^2} + \frac{y^2}{b^2}) d\sigma = \iint_D (\frac{y^2}{a^2} + \frac{x^2}{b^2}) d\sigma$

$\iint_D (\frac{x^2}{a^2} + \frac{y^2}{b^2}) d\sigma = \frac{1}{2} [\text{左端} + \text{右端}]$

$= \frac{1}{2} (\frac{1}{a^2} + \frac{1}{b^2}) \iint_D (x^2 + y^2) d\sigma$

$= \frac{1}{2} (\frac{1}{a^2} + \frac{1}{b^2}) \int_0^{2\pi} d\theta \int_0^R \underline{r^3} dr = \frac{\pi R^4}{4} (\frac{1}{a^2} + \frac{1}{b^2})$



$f(x) \equiv 1$ ✓

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【例3】设区域 $D = \{(x, y) \mid x^2 + y^2 \leq 4, x \geq 0, y \geq 0\}$, $f(x)$ 为 D 上正

值连续函数, a, b 为常数, 则 $\iint_D \frac{a\sqrt{f(x)} + b\sqrt{f(y)}}{\sqrt{f(x)} + \sqrt{f(y)}} d\sigma = \underline{\hspace{2cm}}$.

~~A) $ab\pi$~~ ✓ 4π

~~B) $\frac{ab}{2}\pi$~~ ✓ $\frac{a+b}{2}\pi$

~~C) $(a+b)\pi$~~ ✓ $\frac{a+b}{2}\pi$

~~D) $\frac{a+b}{2}\pi$~~ ✓ $\frac{a+b}{2}\pi$

【解1】直接法

$$\iint_D \frac{a\sqrt{f(x)} + b\sqrt{f(y)}}{\sqrt{f(x)} + \sqrt{f(y)}} d\sigma = \iint_D \frac{a\sqrt{f(y)} + b\sqrt{f(x)}}{\sqrt{f(y)} + \sqrt{f(x)}} d\sigma$$

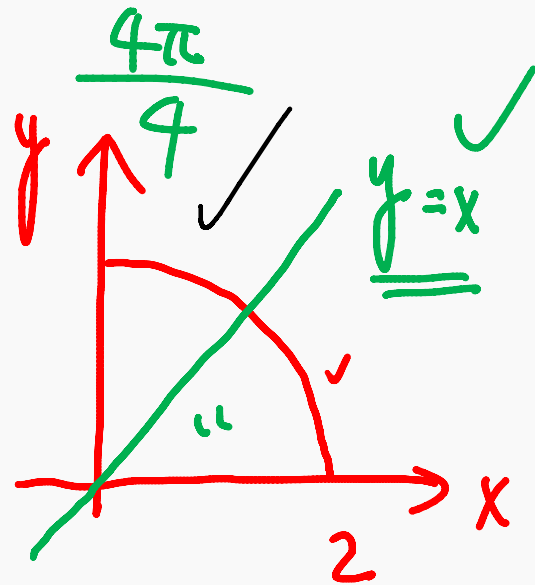
$$\text{原式} = \frac{1}{2} \left[\iint_D \frac{a\sqrt{f(x)} + b\sqrt{f(y)}}{\sqrt{f(y)} + \sqrt{f(y)}} d\sigma + \iint_D \frac{a\sqrt{f(y)} + b\sqrt{f(x)}}{\sqrt{f(y)} + \sqrt{f(x)}} d\sigma \right]$$

$$= \frac{1}{2} \iint_D (a+b) d\sigma = \frac{a+b}{2} \pi$$

【解2】排除法

取 $f(x) \equiv 1$

$$\text{原式} = \frac{1}{2} \iint_D (a+b) d\sigma = \frac{a+b}{2} \pi$$



[特3] $a=b=1$, $a=b=2$

$$\iint_D 1 d\sigma = \pi$$
$$\iint_D 2 d\sigma = 2\pi$$

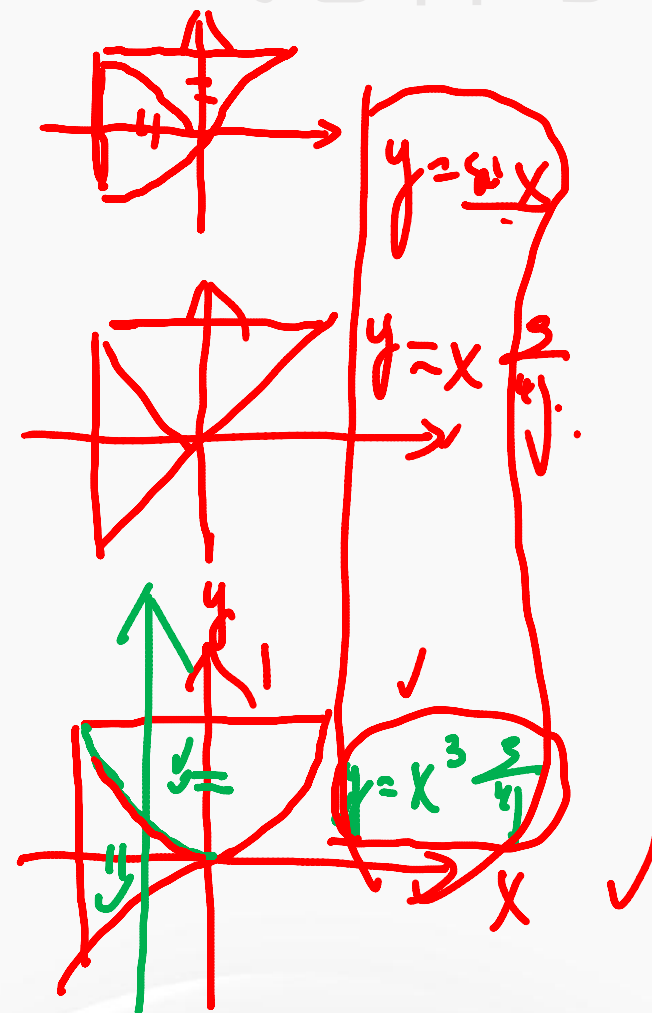
【例4】计算 $\iint_D x[1 + yf(x^2 + y^2)]d\sigma$, 其中 D 是由 $y = x^3, y = 1, x = -1$ 围成的区域, $f(u)$ 为连续函数.

【解】 $\iint_D x[1 + yf(x^2 + y^2)]dxdy$

$$= \iint_D x dxdy + \iint_D xyf(x^2 + y^2) dxdy$$

$\iint_D xyf(x^2 + y^2) dxdy = 0$ (利用奇偶性)

$$\iint_D x dxdy = \int_{-1}^1 dx \int_{x^3}^1 x dy = -\frac{2}{5}$$

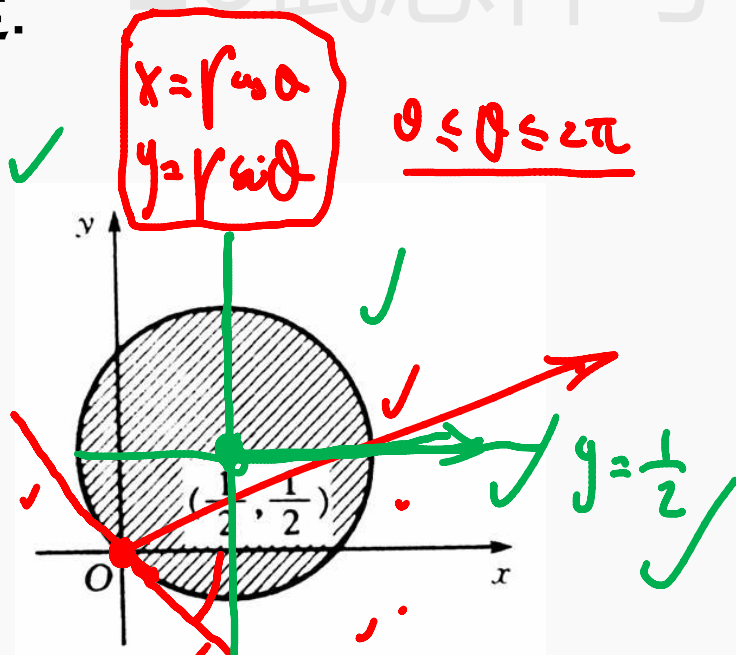


【例7】计算 $\iint_D (x+y) d\sigma$ ，其中 D 由 $x^2 + y^2 \leq x+y$ 所确定。

【解1】 $\iint_D (x+y) d\sigma = \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} d\theta \int_0^{\cos\theta + \sin\theta} (\cos\theta + \sin\theta) r^2 dr$

【解2】* 令 $\begin{cases} x - \frac{1}{2} = r \cos\theta, \\ y - \frac{1}{2} = r \sin\theta, \end{cases}$

$d\sigma = r dr d\theta$



$\iint_D (x+y) d\sigma = \int_0^{2\pi} d\theta \int_0^{\frac{1}{\sqrt{2}}} (r \cos\theta + r \sin\theta + 1) r dr = \int_0^{2\pi} d\theta \int_0^{\frac{1}{\sqrt{2}}} r dr = \frac{\pi}{2}$

【解3】 $\iint_D (x+y) d\sigma = \iint_D [(x-\frac{1}{2}) + (y-\frac{1}{2}) + 1] d\sigma = \frac{\pi}{2}$

【解4】 $\iint_D (x+y) d\sigma = 2 \iint_D x d\sigma = 2 \bar{x} S = 2 \cdot \frac{1}{2} S = \frac{\pi}{2}$

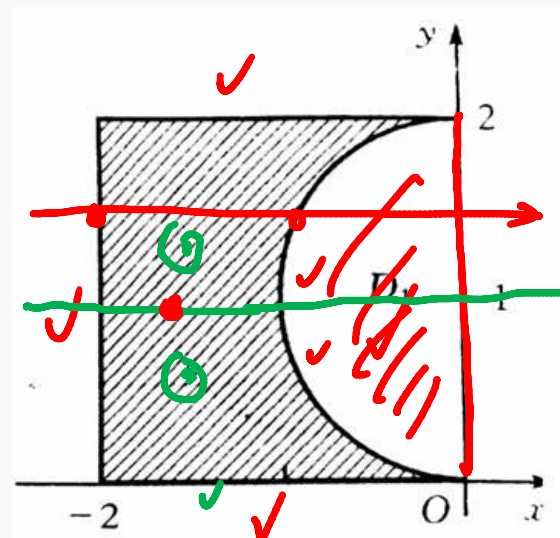
【例8】计算 $\iint_D y dx dy$, 其中 D 是由 $x = -2, y = 0, y = 2$ 以及曲线 $x = -\sqrt{2y - y^2}$ 所围成.

【解1】 $\iint_D y dx dy = \int_0^2 dy \int_{-2}^{-\sqrt{2y-y^2}} y dx$

【解2】 $\iint_D y d\sigma = \int_{-2}^0 dx \int_0^2 y dy - \int_{\frac{\pi}{2}}^{\pi} d\theta \int_0^{2\sin\theta} r^2 \sin\theta dr$

【解3】 $\iint_D y d\sigma = \iint_D [(y-1) + 1] d\sigma = \iint_D d\sigma = 4 - \frac{\pi}{2}$

【解4】 $\iint_D y d\sigma = \bar{y} S = 4 - \frac{\pi}{2}$



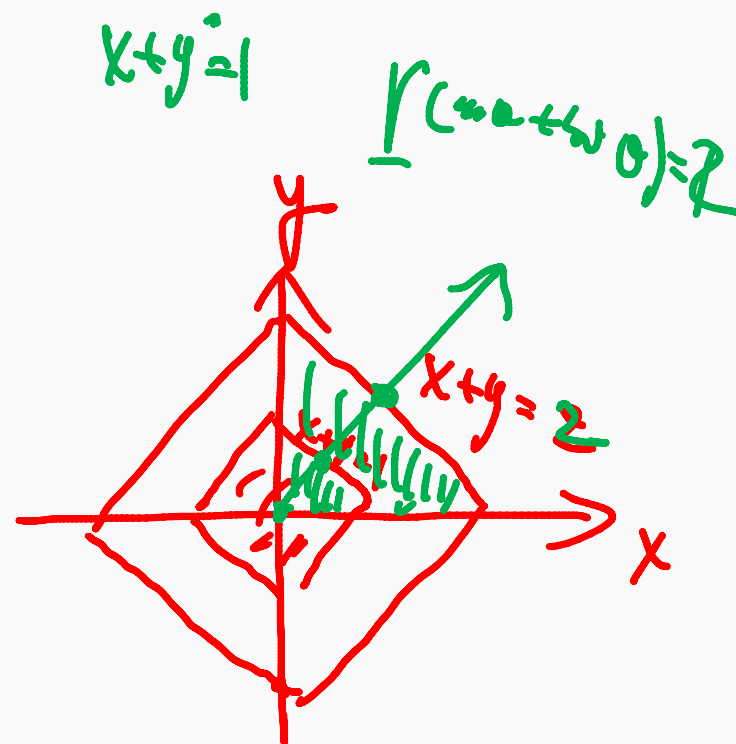
【例9】设二元函数 $f(x, y) = \begin{cases} x^2, & |x| + |y| \leq 1, \\ \frac{1}{\sqrt{x^2 + y^2}}, & 1 < |x| + |y| \leq 2, \end{cases}$

计算二重积分 $\iint_D f(x, y) d\sigma$, 其中 $D = \{(x, y) \mid |x| + |y| \leq 2\}$.

【解】原式 $= 4 \int_0^1 dx \int_0^{1-x} x^2 dy + 4 \int_0^{\frac{\pi}{2}} d\theta \int_{\frac{1}{\sin\theta + \cos\theta}}^2 \frac{2}{\sin\theta + \cos\theta} dr$

$$= \frac{1}{3} + 4 \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sin\theta + \cos\theta}$$

$$= \frac{1}{3} + \frac{4}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sin(\theta + \frac{\pi}{4})} = \frac{1}{3} + 4\sqrt{2} \ln(\sqrt{2} + 1)$$



【例10】

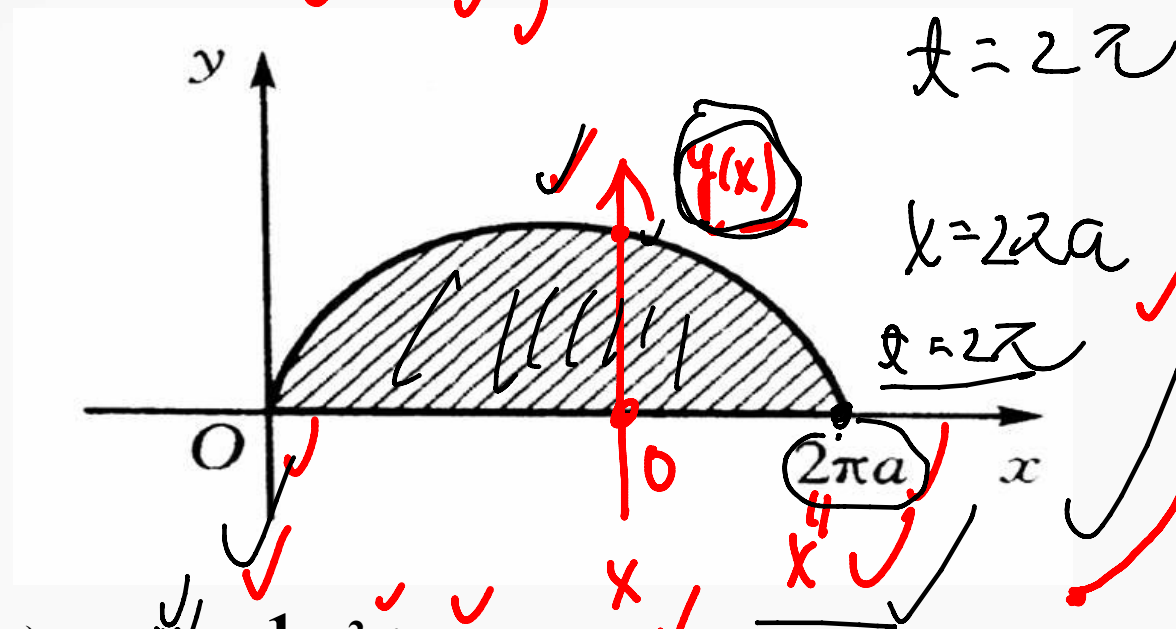
计算

$$\iint_D y^2 d\sigma$$

其中 D 由

$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} (0 \leq t \leq 2\pi)$$

与 $y = 0$ 围成.



【解】

$$\iint_D y^2 d\sigma = \int_0^{2\pi a} dx \int_0^{y(x)} y^2 dy = \frac{1}{3} \int_0^{2\pi a} y^3(x) dx$$

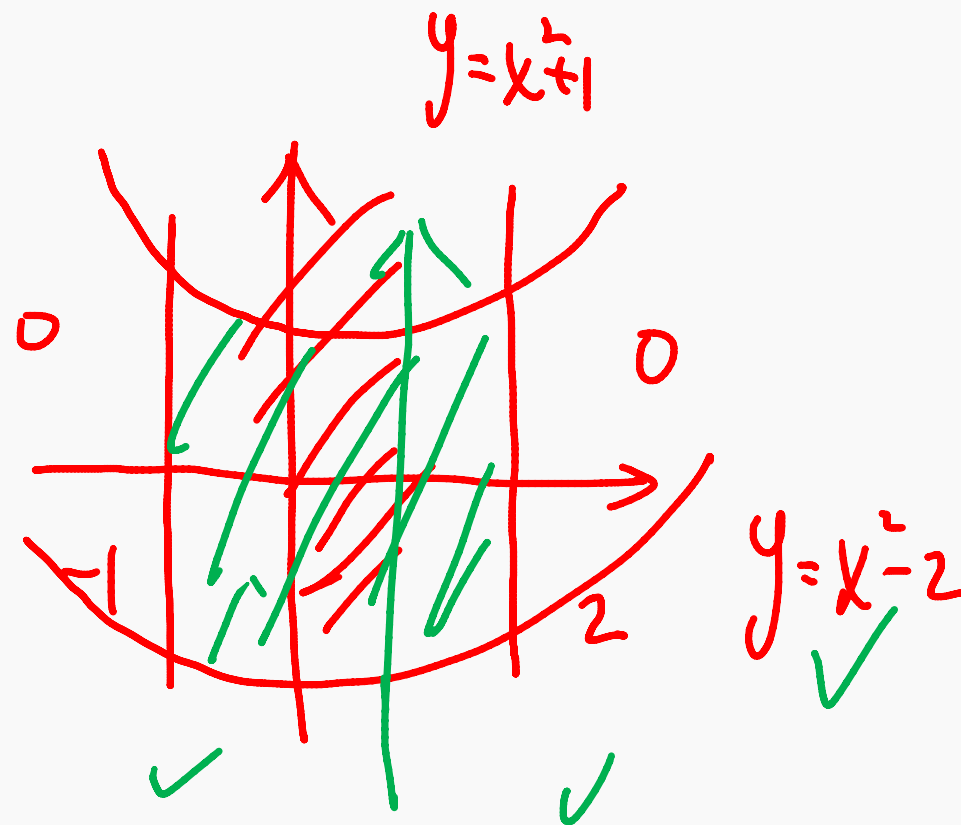
$$= \frac{1}{3} \int_0^{2\pi} a^3 (1 - \cos t)^3 a (1 - \cos t) dt$$

【例11】设 D 是全平面, $f(x) = \begin{cases} x, & -1 \leq x \leq 2 \\ 0, & \text{其它} \end{cases}$

计算 $\iint_D f(x)f(x^2 - y)d\sigma$

【解】原式 $= \int_{-1}^2 dx \int_{x^2-2}^{x^2+1} x(x^2 - y)dy = \frac{9}{4}$

$$\begin{aligned} -1 \leq x^2 - y \leq 2 \\ -1 = x^2 - y \\ y = x^2 + 1 \\ x^2 - y = 2 \\ y = x^2 - 2 \end{aligned}$$



【例12】计算 $\iint_D |x^2 + y^2 - 2y| d\sigma$, 其中 D 由 $x^2 + y^2 \leq 4$ 所确定.

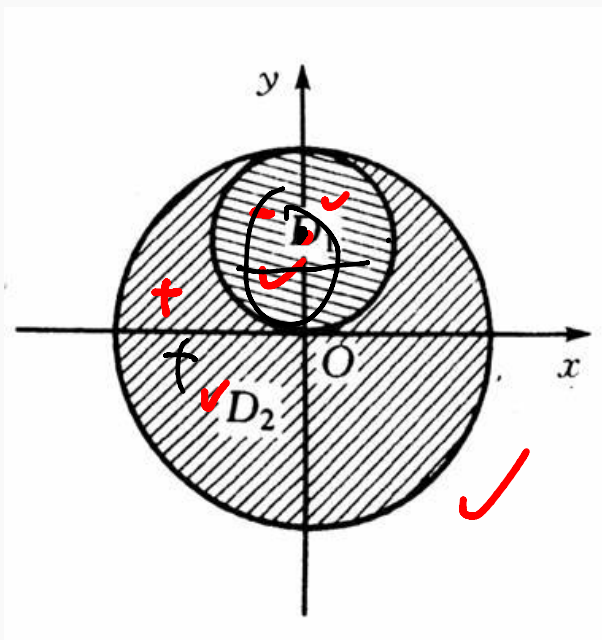
【解】 $\iint_D |x^2 + y^2 - 2y| d\sigma$

$$= \iint_{D_1} (2y - x^2 - y^2) d\sigma + \iint_{D_2} (x^2 + y^2 - 2y) d\sigma$$

$$= \iint_{D_1} (2y - x^2 - y^2) d\sigma + \left[\iint_D (x^2 + y^2 - 2y) d\sigma - \iint_{D_1} (x^2 + y^2 - 2y) d\sigma \right]$$

$$= \iint_D (x^2 + y^2 - 2y) d\sigma + 2 \iint_{D_1} (2y - x^2 - y^2) d\sigma$$

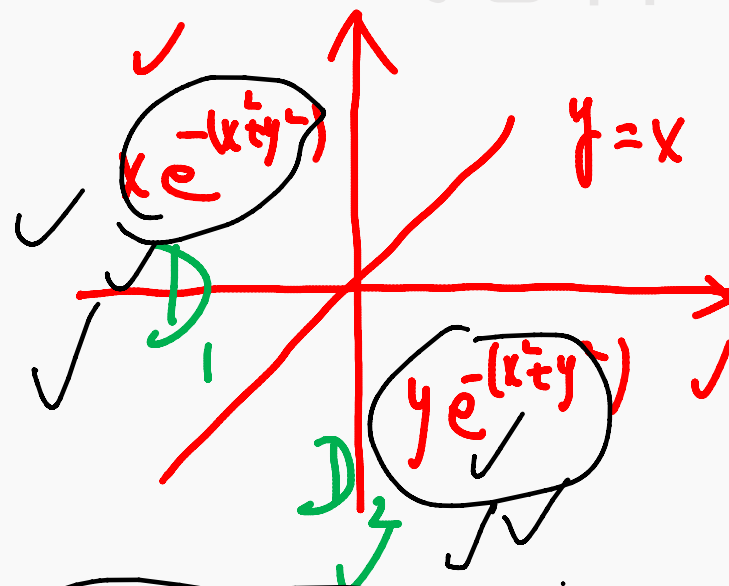
$$= \int_0^{2\pi} d\theta \int_0^2 r^3 dr + 2 \int_0^\pi d\theta \int_0^{2\sin\theta} (2r \sin\theta - r^2) r dr = 9\pi$$



【例13】计算 $\iint_D \min\{x, y\} e^{-(x^2+y^2)} d\sigma$, 其中 D 为全平面.

【解】

$$\begin{aligned} & \iint_D \min\{x, y\} e^{-(x^2+y^2)} d\sigma \\ &= \iint_{D_1} x e^{-(x^2+y^2)} d\sigma + \iint_{D_2} y e^{-(x^2+y^2)} d\sigma \\ & \quad \text{其中 } D_1: y \geq x, D_2: x \geq y \\ &= 2 \int_{-\infty}^{+\infty} dy \int_{-\infty}^y x e^{-x^2} \cdot e^{-y^2} dx \\ &= - \int_{-\infty}^{+\infty} e^{-2y^2} dy \\ & \quad \text{令 } \sqrt{2}y = t, \quad \frac{1}{\sqrt{2}} \int_{-\infty}^{+\infty} e^{-t^2} dt = -\sqrt{\frac{\pi}{2}} \end{aligned}$$



$$\iint_{D(x,y)} f(x,y) dx dy = \iint_{D(y,x)} f(y,x) dy dx$$

【例14】设 $f(x)$ 在区间 $[0,1]$ 上连续, 且 $\int_0^1 f(x)dx = A$

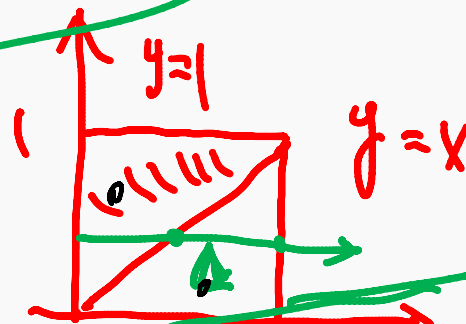
求 $\int_0^1 dx \int_x^1 f(x)f(y)dy$.

【解】 $\int_0^1 dx \int_x^1 f(x)f(y)dy = \int_0^1 dy \int_y^1 f(y)f(x)dx$

$$= \frac{1}{2} \left[\int_0^1 dx \int_x^1 f(x)f(y)dy + \int_0^1 dy \int_y^1 f(x)f(y)dx \right]$$

$$= \frac{1}{2} \iint_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} f(x)f(y)dx dy$$

$$= \frac{1}{2} \int_0^1 f(x)dx \int_0^1 f(y)dy = \frac{A^2}{2}$$



$$\iint_{D(x,y)} f(x,y)dx dy = \iint_{D(y,x)} f(y,x)dy dx$$

10:15

$$= \frac{1}{2} \int_0^1 dx \int_0^1 f(x)f(y)dy$$

$$= \frac{1}{2} \int_0^1 f(x)dx \cdot \int_0^1 f(y)dy$$

题型二 累次积分交换次序及计算

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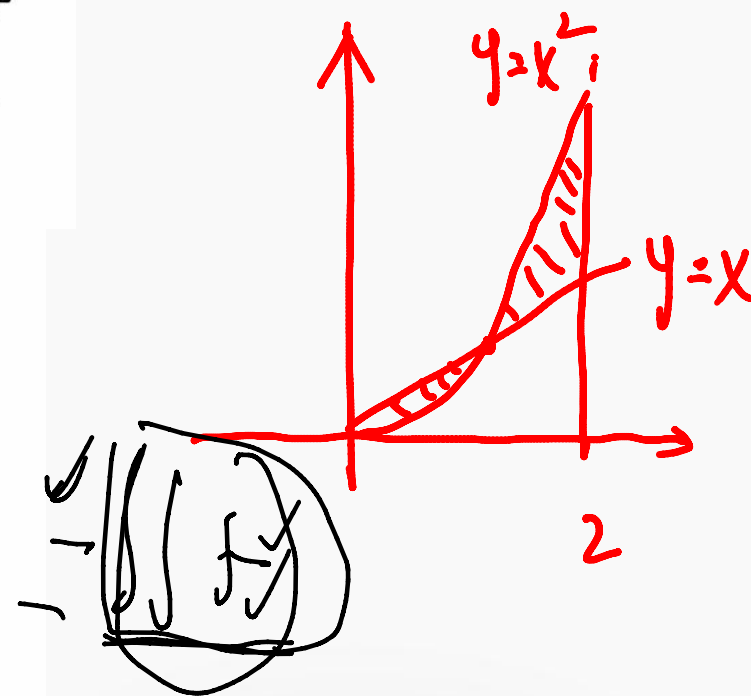
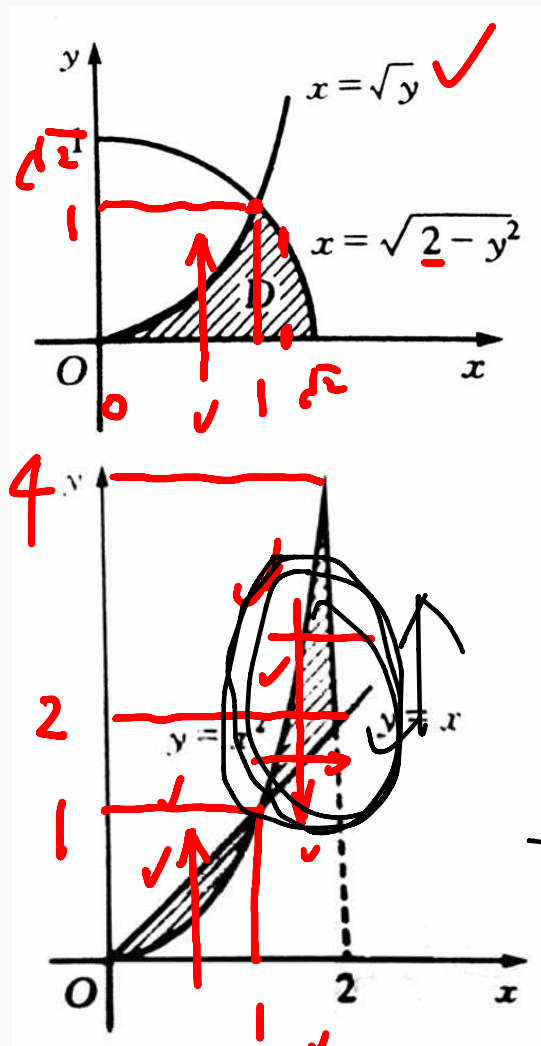
【例1】 交换下列累次积分次序

✓ 1) $I = \int_0^1 dy \int_{\sqrt{y}}^{\sqrt{2-y^2}} f(x, y) dx;$

✓ 3) $I = \int_0^2 dx \int_{x^2}^x f(x, y) dy;$

【解】 1) $I = \int_0^1 dx \int_0^{x^2} f(x, y) dy$
 $+ \int_1^{\sqrt{2}} dx \int_0^{\sqrt{2-x^2}} f(x, y) dy$

3) $I = \int_0^1 dy \int_y^{\sqrt{y}} f(x, y) dx \ominus \int_1^2 dy \int_{\sqrt{y}}^y f(x, y) dx \ominus \int_2^4 dy \int_{\sqrt{y}}^2 f(x, y) dx$



【例2】交换累次积分 $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{2a \cos \theta} f(r \cos \theta, r \sin \theta) r dr$

的次序 ($a > 0$).

$r=0$ $r=2a \cos \theta$ $x^2+y^2=2ax$ $r=2a \cos \theta$

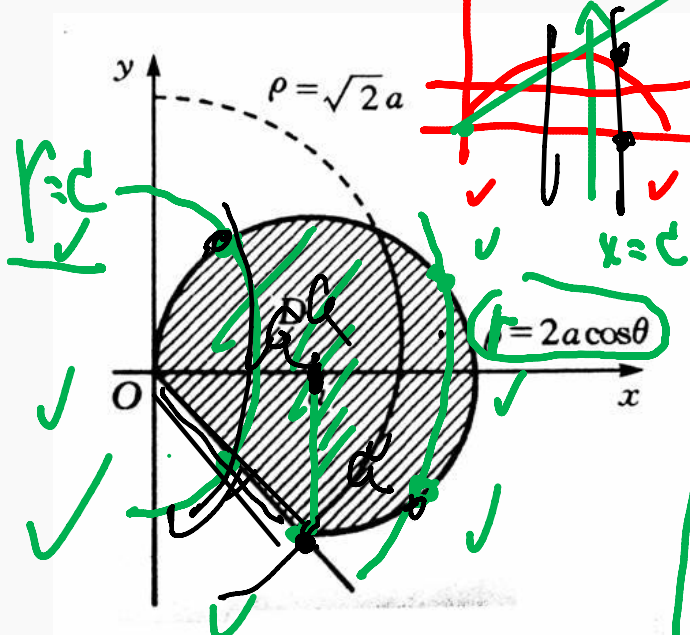
【解1】 $r = 2a \cos \theta$ 是圆 $x^2 + y^2 = 2ax$

$$I = \int_0^{2a} dr \int_{-\arccos \frac{r}{2a}}^{\arccos \frac{r}{2a}} f(r \cos \theta, r \sin \theta) r d\theta$$
$$+ \int_{\sqrt{2}a}^{2a} dr \int_{-\arccos \frac{r}{2a}}^{\arccos \frac{r}{2a}} f(r \cos \theta, r \sin \theta) r d\theta$$

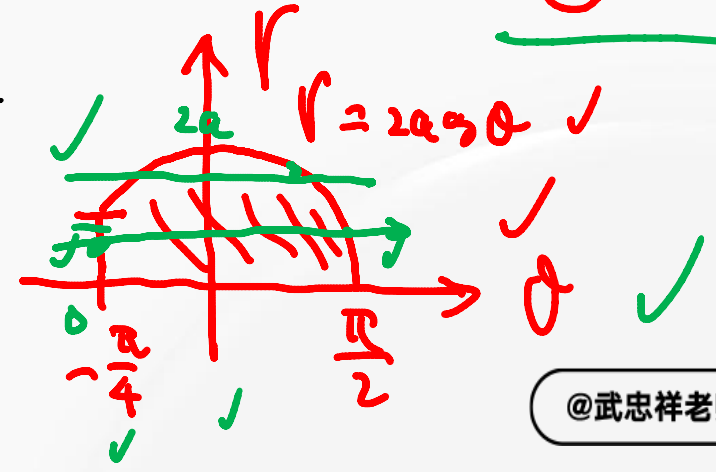
【解2】

$\sin \theta = \frac{r}{2a}$

θ



- ① $\int dx \int f(x,y) dy$ $\frac{x=c}{y=c}$
- ② $\int dy \int f(x,y) dx$ $\frac{y=c}{x=c}$
- ③ $\int d\theta \int f \cdot r dr$ $\frac{\theta=c}{r=c}$
- ④ $\int dr \int f r d\theta$ $\frac{r=c}{\theta=c}$



【例3】累次积分 $\int_0^{\frac{\pi}{2}} d\theta \int_0^{\cos\theta} f(r \cos\theta, r \sin\theta) r dr$ 可写成

$r=0, r'=\cos\theta$

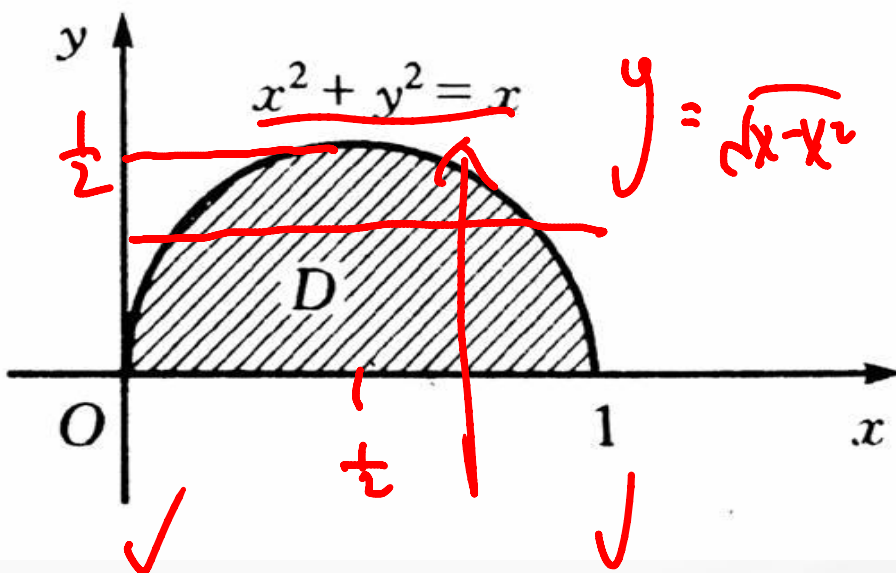
$x^2 + y^2 = x$

A) $\int_0^1 dy \int_0^{\sqrt{y-y^2}} f(x, y) dx,$

B) $\int_0^1 dy \int_0^{\sqrt{1-y^2}} f(x, y) dx,$

C) $\int_0^1 dx \int_0^1 f(x, y) dy,$

D) $\int_0^1 dx \int_0^{\sqrt{x-x^2}} f(x, y) dy.$



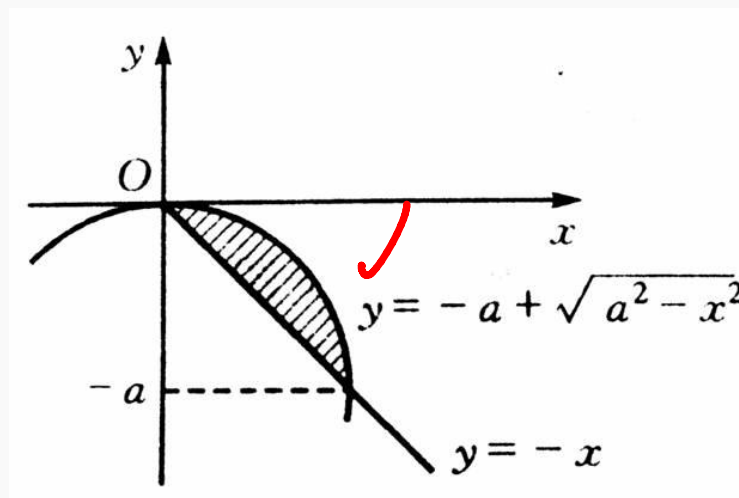
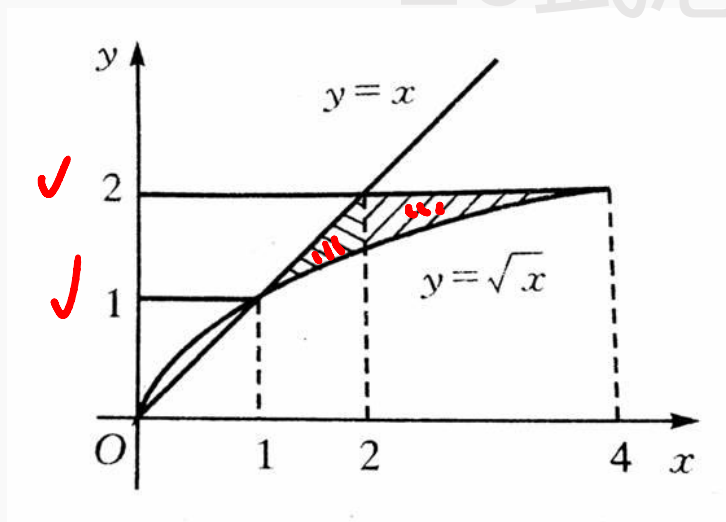
【例4】 计算下列累次积分

2) $\int_1^2 dx \int_{\sqrt{x}}^x \sin \frac{\pi x}{2y} dy + \int_2^4 dx \int_{\sqrt{x}}^2 \sin \frac{\pi x}{2y} dy;$

3) $\int_0^a dx \int_{-x}^{-a+\sqrt{a^2-x^2}} \frac{1}{\sqrt{4a^2-(x^2+y^2)}} dy; (a > 0)$

【解】 2) 原式 = $\int_1^2 dy \int_y^{y^2} \sin \frac{\pi x}{2y} dx$

3) $\int_{-\frac{\pi}{4}}^0 d\theta \int_0^{-2a \sin \theta} \frac{r}{\sqrt{4a^2-r^2}} dr$



【例5】设 $f(x)$ 为连续. 证明: $\iint_D f(x-y) dx dy = \int_{-A}^A f(t)(A-|t|) dt$

$$D: |x| \leq \frac{A}{2}, |y| \leq \frac{A}{2}$$

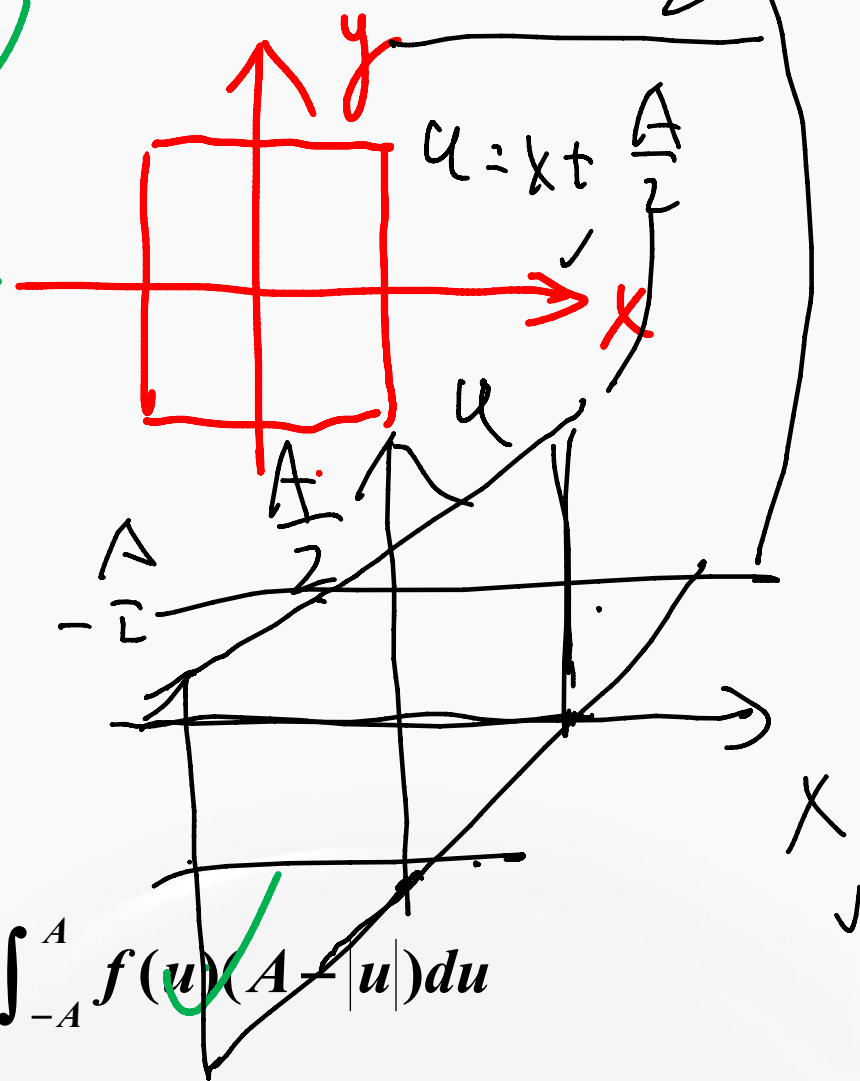
【证】 $\iint_D f(x-y) dx dy = \int_{-\frac{A}{2}}^{\frac{A}{2}} dx \int_{-\frac{A}{2}}^{\frac{A}{2}} f(x-y) dy$

$$\int_{-\frac{A}{2}}^{\frac{A}{2}} f(x-y) dy = \int_{x-\frac{A}{2}}^{x+\frac{A}{2}} f(u) du \quad (x-y=u)$$

$$\iint_D f(x-y) dx dy = \int_{-\frac{A}{2}}^{\frac{A}{2}} dx \int_{x-\frac{A}{2}}^{x+\frac{A}{2}} f(u) du$$

$$= \int_{-A}^0 du \int_{-\frac{A}{2}}^{u+\frac{A}{2}} f(u) dx + \int_0^A du \int_{u-\frac{A}{2}}^{\frac{A}{2}} f(u) dx$$

$$= \int_{-A}^0 f(u)(A+u) du + \int_0^A f(u)(A-u) du = \int_{-A}^A f(u)(A-|u|) du$$



题型三 与二重积分有关的综合题

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【例1】设 $f(x)$ 为连续函数, $F(t) = \int_1^t dy \int_y^t f(x) dx$, 则 $F'(2)$ 等于

- A) $2f(2)$ B) $f(2)$ C) $-f(2)$ D) 0

【解1】 $F(t) = \int_1^t dx \int_1^x f(x) dy = \int_1^t (x-1)f(x) dx$

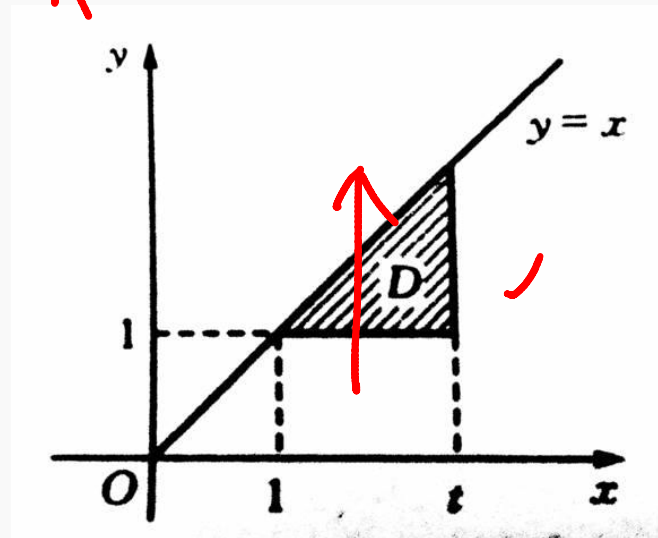
$$F'(t) = (t-1)f(t)$$

$$F'(2) = f(2)$$

故应选 (B)

【解2】排除法 $f(t) = 1$

$$F(t) = \int_1^t (x-y) dy$$



【例2】 设区域 D 由 $x^2 + y^2 \leq y$ 和 $x \geq 0$ 所确定, $f(x, y)$ 为 D

上的连续函数, 且 $f(x, y) = \sqrt{1 - x^2 - y^2} - \frac{8}{\pi} \iint_D f(u, v) du dv$, 求 $f(x, y)$.

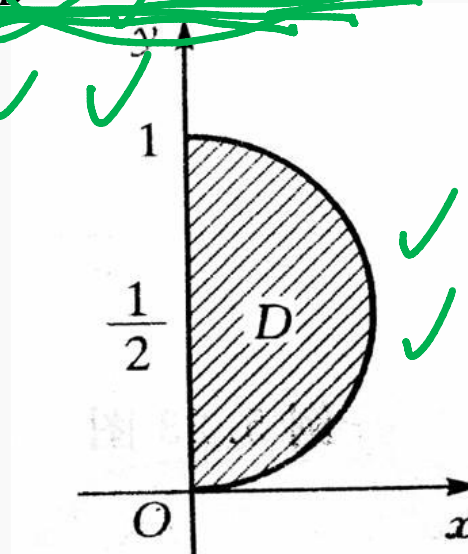
【解1】 令 $\iint_D f(u, v) du dv = A$,

则 $f(x, y) = \sqrt{1 - x^2 - y^2} - \frac{8}{\pi} A$

$$\iint_D \left[\sqrt{1 - x^2 - y^2} - \frac{8}{\pi} A \right] dx dy = A$$

$$A = \frac{1}{2} \iint_D \sqrt{1 - x^2 - y^2} dx dy = \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sin \theta} \sqrt{1 - r^2} r dr = \frac{1}{6} \left(\frac{\pi}{2} - \frac{2}{3} \right)$$

【解2】 $\iint_D f(x, y) dx dy = \iint_D \sqrt{1 - x^2 - y^2} dx dy - \iint_D f(u, v) du dv$



$$\frac{1}{2} \pi \left(\frac{1}{2} \right)^2 = \frac{\pi}{8}$$

【例3】设 $f(t)$ 在 $[0, +\infty)$ 上连续, 且满足

$$f(t) = e^{4\pi^2} + \iint_{x^2+y^2 \leq 4t^2} f\left(\frac{1}{2}\sqrt{x^2+y^2}\right) dx dy \quad \text{求 } f(t).$$

【解】 $\iint_{x^2+y^2 \leq 4t^2} f\left(\frac{1}{2}\sqrt{x^2+y^2}\right) dx dy = \int_0^{2\pi} d\theta \int_0^{2t} f\left(\frac{1}{2}r\right) r dr$

$$= 2\pi \int_0^{2t} r f\left(\frac{1}{2}r\right) dr$$

$$f(t) = e^{4\pi^2} + 2\pi \int_0^{2t} r f\left(\frac{1}{2}r\right) dr$$

$$f'(t) = 8\pi t e^{4\pi^2} + 8\pi t f(t)$$

$$f(t) = e^{\int 8\pi t dt} \left[\int 8\pi t e^{-\int 8\pi t dt} dt + C \right] = (4\pi^2 + C) e^{4\pi^2}$$

由 $f(0) = 1$ 得 $C = 1$, $f(t) = (4\pi^2 + 1) e^{4\pi^2}$

$$f(t) = e^{4\pi^2} + \int_0^t f(x) dx$$

$$\int_0^1 dx \int_0^k f(y) dy$$

【例4】设 $f(x, y)$ 是定义在 $0 \leq x \leq 1, 0 \leq y \leq 1$ 上的连续函数,

$f(0,0) = -1$, 求极限 $\lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} dt \int_x^{\sqrt{t}} f(t, u) du}{1 - e^{-x^3}}$.

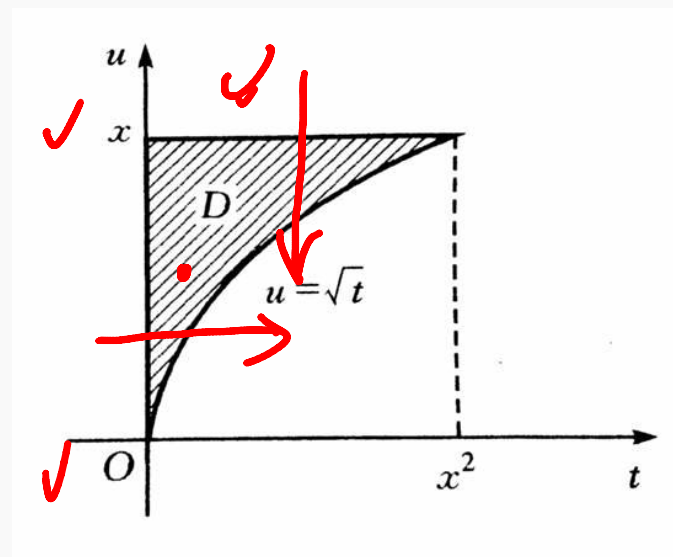
【解1】 $\lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} dt \int_x^{\sqrt{t}} f(t, u) du}{1 - e^{-x^3}} = \lim_{x \rightarrow 0^+} \frac{-\int_0^{x^2} du \int_0^u f(t, u) dt}{x^3}$

(*)

$$= -\lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} f(t, x) dt}{3x^2} = -\lim_{x \rightarrow 0^+} \frac{x^2 f(c, x)}{3x^2} = -\frac{1}{3} f(0,0) = \frac{1}{3}$$

【解2】 $\int_0^{x^2} dt \int_x^{\sqrt{t}} f(t, u) du = -\iint_D f(t, u) dt du = -f(\xi, \eta) S$

$$S = \int_0^{x^2} dt \int_{\sqrt{t}}^x du = \int_0^{x^2} (x - \sqrt{t}) dt = \frac{1}{3} x^3$$



$$\left(\int_0^x \left(\int_0^u f(t, u) dt \right) du \right) = \bar{F}(u) = \bar{F}(x)$$

【例5】 设 $f(x, y)$ 在单位圆 $x^2 + y^2 \leq 1$ 上有连续一阶偏导数，

且在边界上取值为零，证明： $f(0,0) = \lim_{\varepsilon \rightarrow 0^+} \frac{-1}{2\pi} \iint_D \frac{xf_x + yf_y}{x^2 + y^2} dx dy$

其中 D 为圆环域 $\varepsilon^2 \leq x^2 + y^2 \leq 1$.

【证】

$$\iint_D \frac{xf_x + yf_y}{x^2 + y^2} dx dy$$

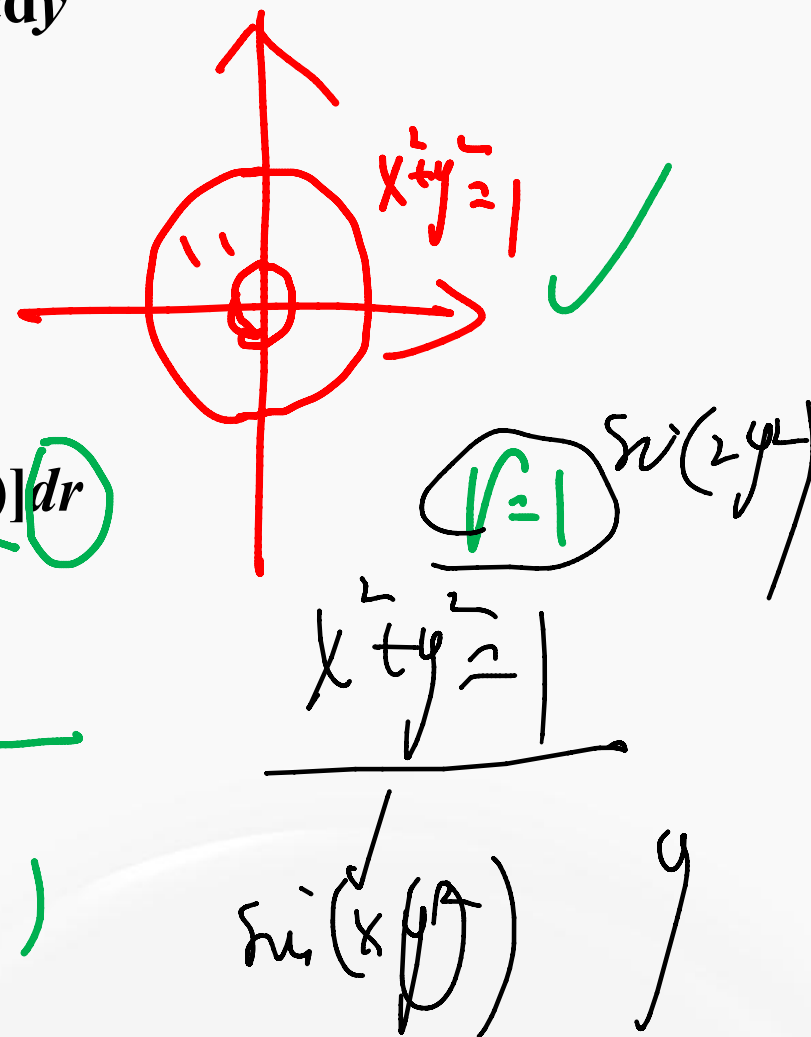
$$= \int_0^{2\pi} d\theta \int_{\varepsilon}^1 [\cos \theta f_x(r \cos \theta, r \sin \theta) + \sin \theta f_y(r \cos \theta, r \sin \theta)] dr$$

$$= \int_0^{2\pi} [f(r \cos \theta, r \sin \theta)]_{\varepsilon}^1 d\theta = - \int_0^{2\pi} f(\varepsilon \cos \theta, \varepsilon \sin \theta) d\theta$$

$$= -2\pi f(\varepsilon \cos \bar{\theta}, \varepsilon \sin \bar{\theta}), \bar{\theta} \in [0, 2\pi]$$

$$\lim_{\varepsilon \rightarrow 0^+} \frac{-1}{2\pi} \iint_D \frac{xf_x + yf_y}{x^2 + y^2} dx dy = \lim_{\varepsilon \rightarrow 0^+} f(\varepsilon \cos \bar{\theta}, \varepsilon \sin \bar{\theta}) = f(0,0)$$

$f_x(x, y)$



$|f''(x)| \leq M, \quad \left| \int_a^b f(x) dx \right| \leq (b-a)M$

【例6】 设二元函数 $f(x, y)$ 在平面域 $D: 0 \leq x \leq 1, 0 \leq y \leq 1$

上有二阶连续偏导数, 在 D 的边界上取值为零, 且在

D 上有 $\left| \frac{\partial^2 f}{\partial x \partial y} \right| \leq M$, 试证 $\left| \iint_D f(x, y) dx dy \right| \leq \frac{M}{4}$.

【证1】 由题设知 $f(0, y) = 0, f(1, y) = 0, f(x, 0) = 0, f(x, 1) = 0$.

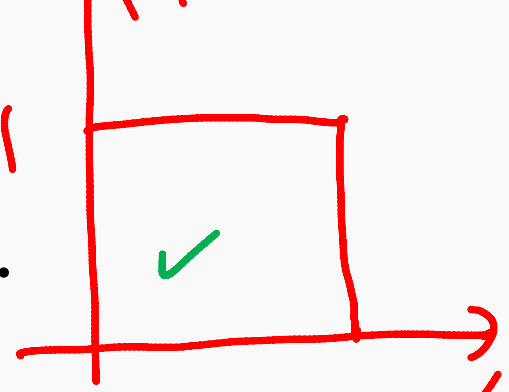
从而有 $f_y(0, y) = 0, f_y(1, y) = 0, f_x(x, 0) = 0, f_x(x, 1) = 0$.

$$\begin{aligned} \iint_D f(x, y) dx dy &= \int_0^1 dx \left[\int_0^1 f(x, y) dy \right] \\ &= \int_0^1 \left[y f(x, y) \Big|_0^1 - \int_0^1 y \frac{\partial f}{\partial y} dy \right] dx = \int_0^1 y dy \int_0^1 \frac{\partial f}{\partial y} d(1-x) \end{aligned}$$

$$= \int_0^1 y \left[(1-x) \left(\frac{\partial f}{\partial y} \right) \Big|_0^1 - \int_0^1 (1-x) \frac{\partial^2 f}{\partial x \partial y} dx \right] dy = - \iint_D y(1-x) \frac{\partial^2 f}{\partial x \partial y} dx dy$$

则 $\left| \iint_D f(x, y) dx dy \right| \leq \iint_D y(1-x) \left| \frac{\partial^2 f}{\partial x \partial y} \right| dx dy \leq M \iint_D y(1-x) dx dy = \frac{M}{4}$

① 泰勒
② 分部



$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$

$\int f(x, y) dy$

$y f(x, y) \Big|_0^1 - \int_0^1 y \frac{\partial f}{\partial y} dy$



【例6】设二元函数 $f(x, y)$ 在平面域 $D: 0 \leq x \leq 1, 0 \leq y \leq 1$

上有二阶连续偏导数，在 D 的边界上取值为零，且在

$$D \text{ 上有 } \left| \frac{\partial^2 f}{\partial x \partial y} \right| \leq M, \text{ 试证 } \left| \iint_D f(x, y) dx dy \right| \leq \frac{M}{4}.$$

【证2】 $f(x, y) = f(x, y) - f(0, y)$

$$(f(0, y) = 0)$$

$$= x f_x(\xi, y)$$

(拉格朗日中值定理)

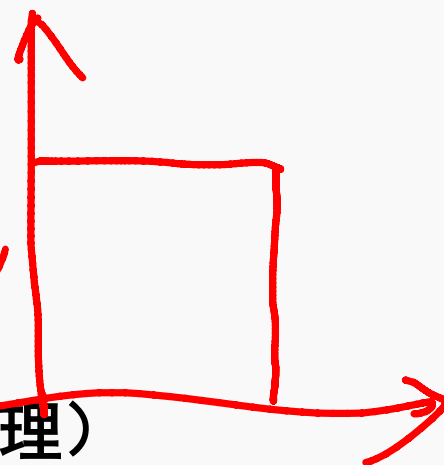
$$= x [f_x(\xi, y) - f_x(\xi, 0)]$$

$$(f_x(x, 0) = 0)$$

$$= xy f_{xy}(\xi, \eta)$$

(拉格朗日中值定理)

$$\left| \iint_D f(x, y) dx dy \right| \leq \iint_D |xy f_{xy}(\xi, \eta)| dx dy \leq M \iint_D xy dx dy = \frac{M}{4}.$$



题型四 与二重积分有关的积分不等式问题

【例1】 设 $I_1 = \iint_D \cos \sqrt{x^2 + y^2} d\sigma$, $I_2 = \iint_D \cos(x^2 + y^2) d\sigma$, $I_3 = \iint_D \cos(x^2 + y^2)^2 d\sigma$

其中 $D = \{(x, y) | x^2 + y^2 \leq 1\}$, 则

A) $I_3 > I_2 > I_1$

B) $I_1 > I_2 > I_3$

C) $I_2 > I_1 > I_3$

D) $I_3 > I_1 > I_2$

【解】 $\sqrt{x^2 + y^2} \geq x^2 + y^2 \geq (x^2 + y^2)^2$

$$\cos \sqrt{x^2 + y^2} \leq \cos(x^2 + y^2) \leq \cos(x^2 + y^2)^2$$



【例2】 设

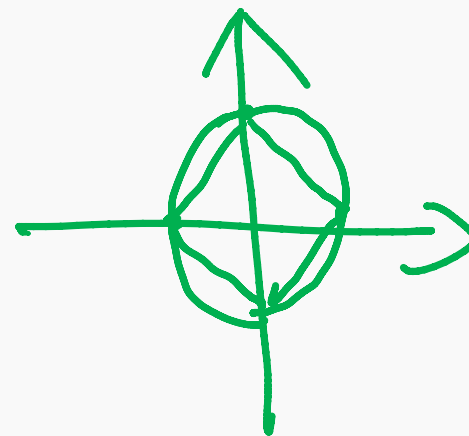
$$I_1 = \iint_{x^2+y^2 \leq 1} (x^2+y^2) d\sigma, I_2 = \iint_{|x|+|y| \leq 1} 2|xy| d\sigma, I_3 = \iint_{|x|+|y| \leq 1} (x^2+y^2) d\sigma, \text{ 则}$$

A) $I_2 < I_3 < I_1$;

B) $I_1 < I_2 < I_3$;

C) $I_3 < I_1 < I_2$;

D) $I_3 < I_2 < I_1$;



【例3】 设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(x) > 0$,

证明: $\int_a^b f(x) dx \int_a^b \frac{1}{f(x)} dx \geq (b-a)^2$.

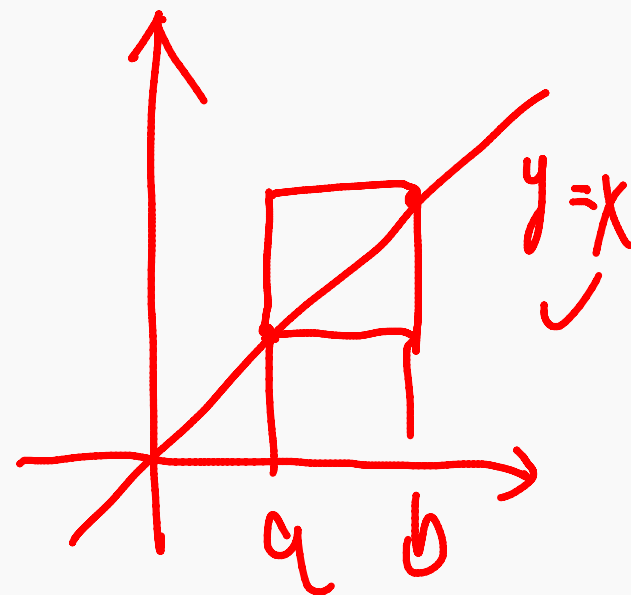
【证1】 $D = \{(x, y) \mid a \leq x \leq b, a \leq y \leq b\}$

① $\int_a^b f(x) dx \cdot \int_a^b \frac{1}{f(x)} dx = \int_a^b f(x) dx \cdot \int_a^b \frac{1}{f(y)} dy = \iint_D \frac{f(x)}{f(y)} dx dy$

② $\int_a^b f(x) dx \cdot \int_a^b \frac{1}{f(x)} dx = \frac{1}{2} \left[\iint_D \frac{f(x)}{f(y)} dx dy + \iint_D \frac{f(y)}{f(x)} dx dy \right]$

$= \frac{1}{2} \iint_D \frac{f^2(x) + f^2(y)}{f(x)f(y)} dx dy = \iint_D \frac{f^2(x) + f^2(y)}{2f(x)f(y)} dx dy \geq \iint_D 1 dx dy$

$= (b-a)^2$



【例3】设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(x) > 0$,

证明: $\int_a^b f(x) dx \int_a^b \frac{1}{f(x)} dx \geq (b-a)^2.$

【证2】由柯西积分不等式得

$$\left(\int_a^b f g dx \right)^2 \leq \int_a^b f^2 dx \cdot \int_a^b g^2 dx$$

$$\begin{aligned} \int_a^b f(x) dx \int_a^b \frac{1}{f(x)} dx &= \int_a^b (\sqrt{f(x)})^2 dx \cdot \int_a^b \left(\frac{1}{\sqrt{f(x)}} \right)^2 dx \\ &\geq \left(\int_a^b \sqrt{f(x)} \cdot \frac{1}{\sqrt{f(x)}} dx \right)^2 = (b-a)^2 \end{aligned}$$

【例4】设 $f(x)$ 在 $[0,1]$ 上单调减的正值函数, 证明: $\frac{\int_0^1 xf^2(x)dx}{\int_0^1 xf(x)dx} \leq \frac{\int_0^1 f^2(x)dx}{\int_0^1 f(x)dx}$

【证1】 $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$

$$I = \int_0^1 f^2(x)dx \int_0^1 xf(x)dx - \int_0^1 xf^2(x)dx \int_0^1 f(x)dx$$

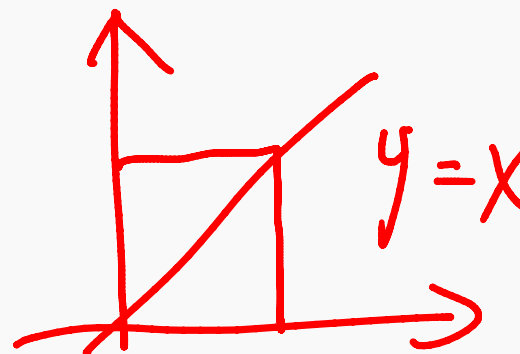
$$\textcircled{1} = \int_0^1 f^2(x)dx \cdot \int_0^1 yf(y)dy - \int_0^1 xf^2(x)dx \cdot \int_0^1 f(y)dy$$

$$= \iint_D yf^2(x)f(y)dxdy - \iint_D xf^2(x)f(y)dxdy = \iint_D f^2(x)f(y)(y-x)dxdy$$

$$I = \iint_D f^2(x)f(y)(y-x)dxdy = \iint_D f^2(y)f(x)(x-y)dxdy$$

$$\textcircled{2} I = \frac{1}{2} \left[\iint_D f^2(x)f(y)(y-x)dxdy + \iint_D f^2(y)f(x)(x-y)dxdy \right]$$

$$= \frac{1}{2} \iint_D f(x)f(y)(y-x)[f(x)-f(y)]dxdy \geq 0$$



2

一.复习消化强化课内容 (例题, 学习包习题)

二.严选题

三.660题

四.330题



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你就慢了



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